Ques 1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Ans:

Basic Filter

It is a Simple filtering (We only select values which i want to see)

Example: Show only **Year = 2023**

Use when: quick filtering

Advanced Filter

when we use conditions like:

greater than , contains , between Then We used Advanced Filter

Example:

Sales > 50,000

Use when: condition-based filtering

Slicer

A Slicer is a visual filter placed on the report page that users can interact with

Example:

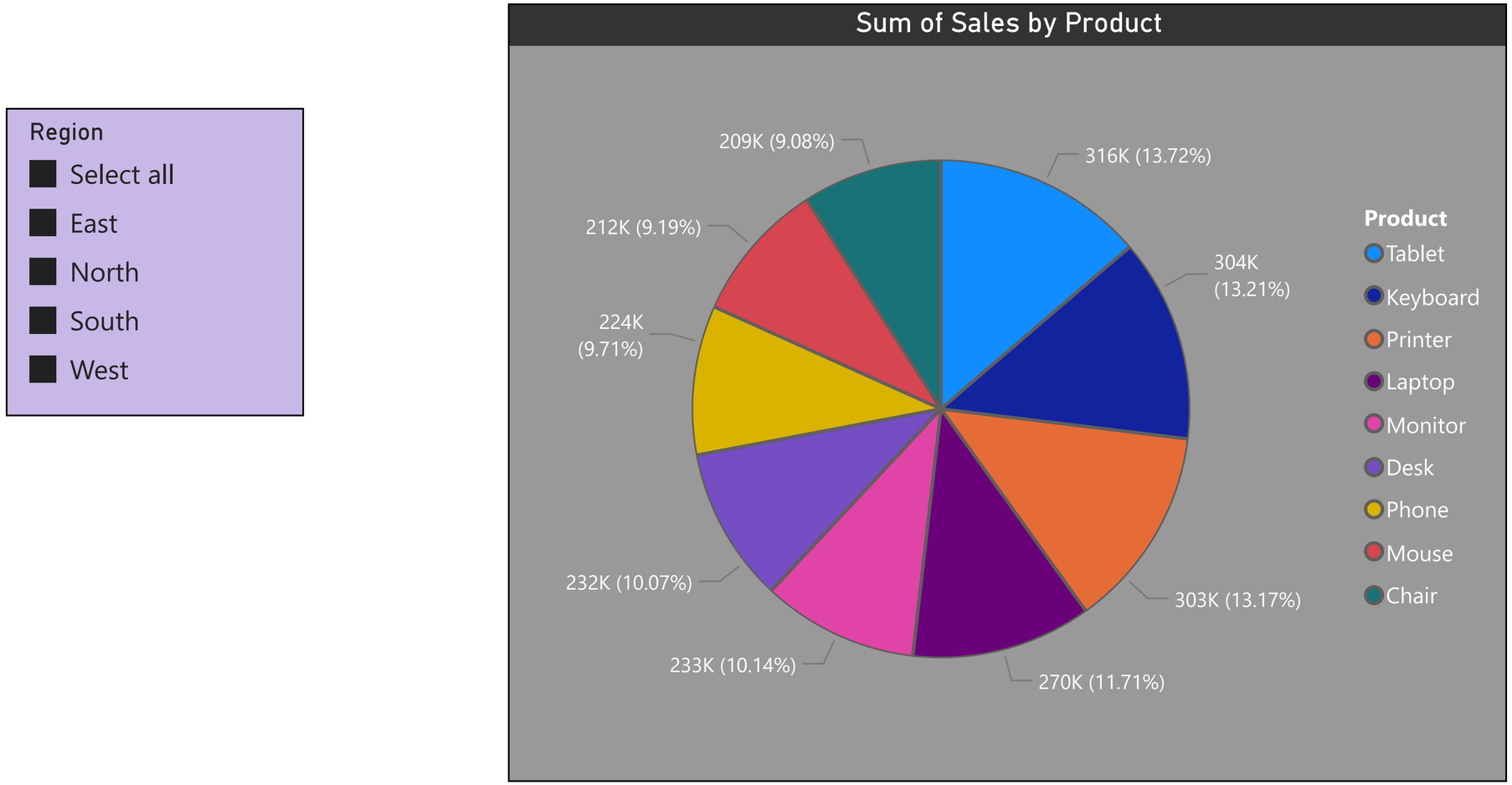
User clicks Region = "India"

Use when: **interactive dashboard**

Ques 2: Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

Date	Product	Category	Region	Sum of S
Tuesday, May 30, 2023	Keyboard	Accessories	North	
Saturday, October 30, 2021	Keyboard	Accessories	South	
Sunday, October 15, 2023	Keyboard	Accessories	East	
Tuesday, March 30, 2021	Keyboard	Accessories	West	
Tuesday, April 25, 2023	Monitor	Accessories	West	
Saturday, September 25, 2021	Monitor	Accessories	North	
Thursday, September 28, 2023	Monitor	Accessories	South	
Thursday, February 25, 2021	Monitor	Accessories	East	
Monday, March 20, 2023	Tablet	Electronics	East	
Friday, August 20, 2021	Tablet	Electronics	West	
Tuesday, August 22, 2023	Tablet	Electronics	North	
Thursday, January 14, 2021	Tablet	Electronics	South	
Friday, December 01, 2023	Tablet	Electronics	North	
Wednesday, February 15, 2023	Phone	Electronics	South	
Friday, July 15, 2022	Phone	Electronics	East	
Wednesday, August 16, 2023	Tablet	Electronics	East	
Tuesday, July 18, 2023	Phone	Electronics	West	
Tuesday, December 20, 2022	Phone	Electronics	North	
Friday, October 25, 2024	Laptop	Electronics	East	1
Total				**2302**

Ques 3: Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region



Ques 4: What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

Answer:

Visual-Level Filter

A Visual-Level Filter affects only one specific visual (chart or table) in a report.

Example

Suppose we have:

- Pie Chart → Sales by Product
- Bar Chart → Sales by Region

If we apply a Visual-Level Filter to the Pie Chart for Year = 2023, then:

- Pie Chart → shows only 2023 data
- Bar Chart → still shows all years

So the filter affects only that visual.

Where it appears

Right side Filters Pane → Filters on this visual.

Page-Level Filter

A Page-Level Filter affects all visuals on a single report page.

Example

Ques5:Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India

Country	Product	Sum of Sales
Dubai	Keyboard	175
Dubai	Monitor	350
Dubai	Tablet	650
India	Chair	208996
India	Desk	231793
India	Keyboard	303877
India	Laptop	269552
India	Monitor	233033
India	Mouse	211644
India	Phone	223560
India	Printer	303220
India	Tablet	315168
Total		**2302018**

Ques 6 : What is the role of Drill-Down in Power BI? How does it improve data exploration?

Answer:

Drill-Down is a feature in Power BI that allows users to explore data at different levels of detail within the same visual.

It helps users move from high-level summary data to more detailed data.

Role of Drill-Down

Drill-down helps users:

Analyze data step-by-step

Identify trends and patterns

Understand detailed information behind summary data

Drill-down improves analysis because:

It allows users to navigate through different data levels.

It helps identify which category contributes the most.

It makes reports interactive and easy to explore.

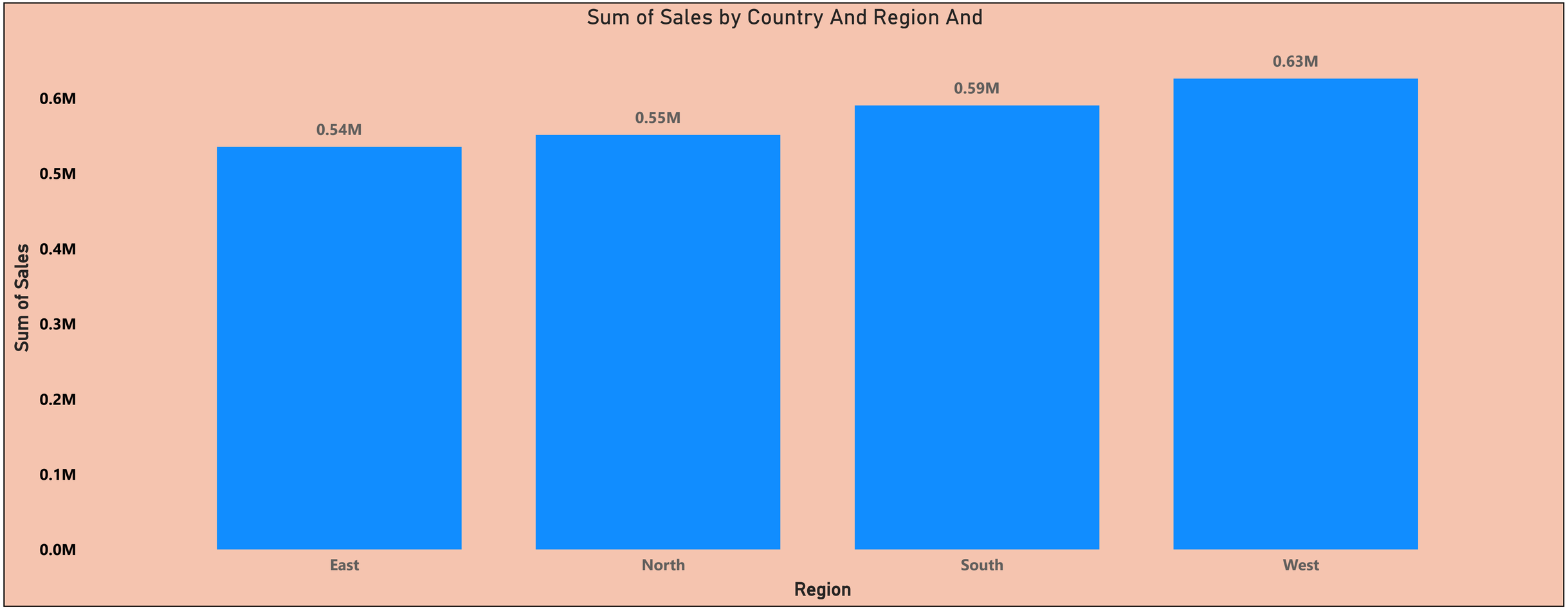
It helps find hidden insights in the data.

Simple Real-Life Example

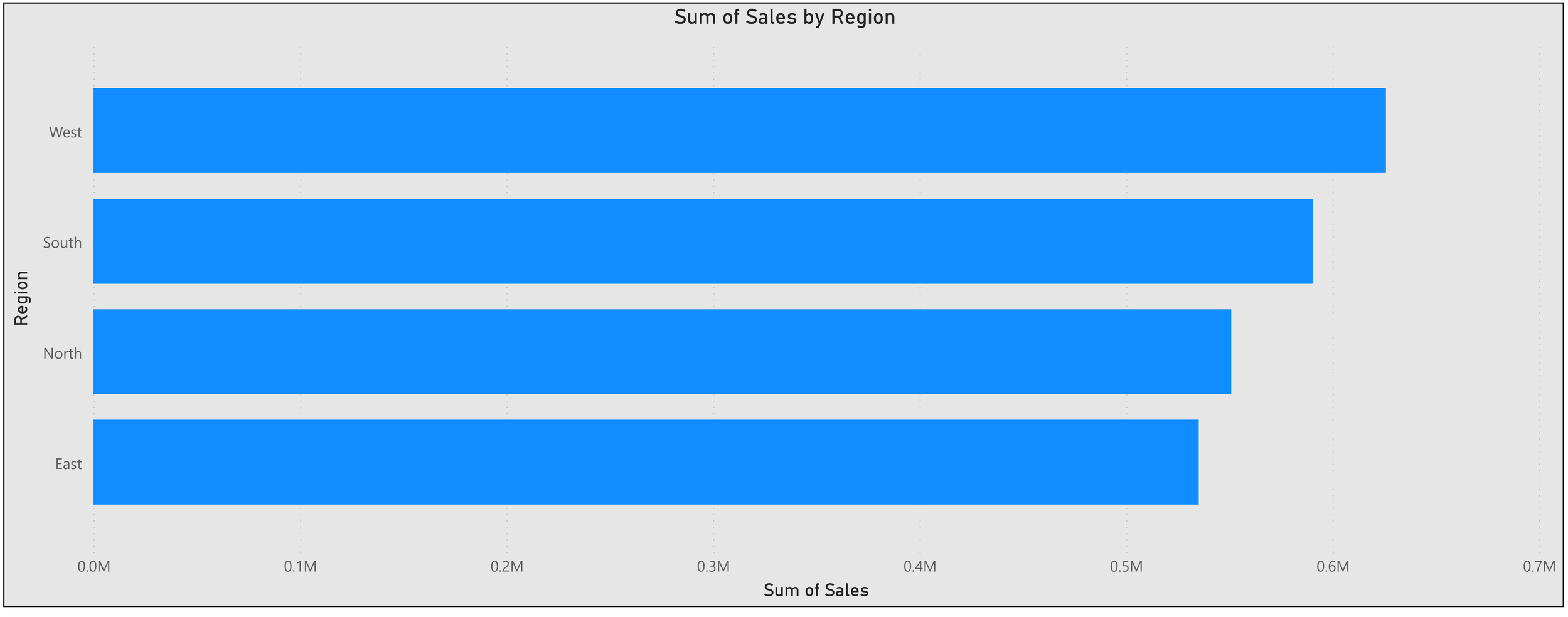
Imagine a company sales dashboard.

First you see:

Ques 7 : Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.

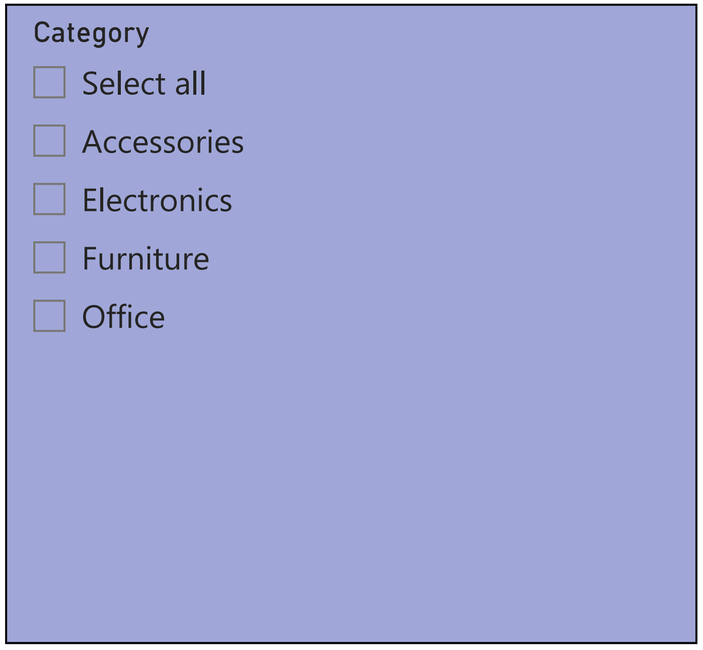


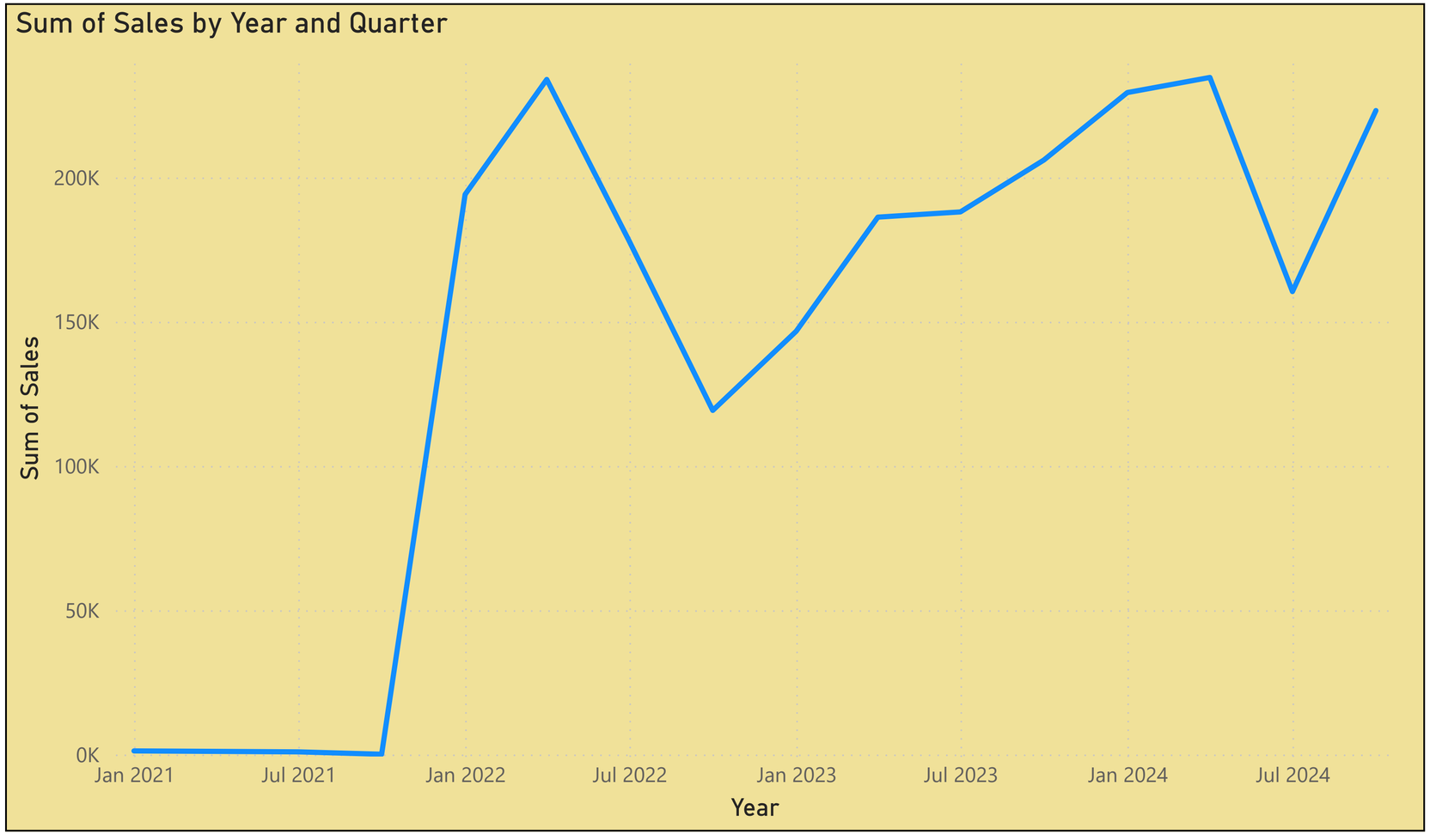
Ques 8 :Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region wise sales.



Region	Sum of Sales
East	535112
North	550844
South	590265
West	625797
Total	**2302018**

Ques 9: Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy





Ques 10 : Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

Answer:

Drill-Down in Power BI

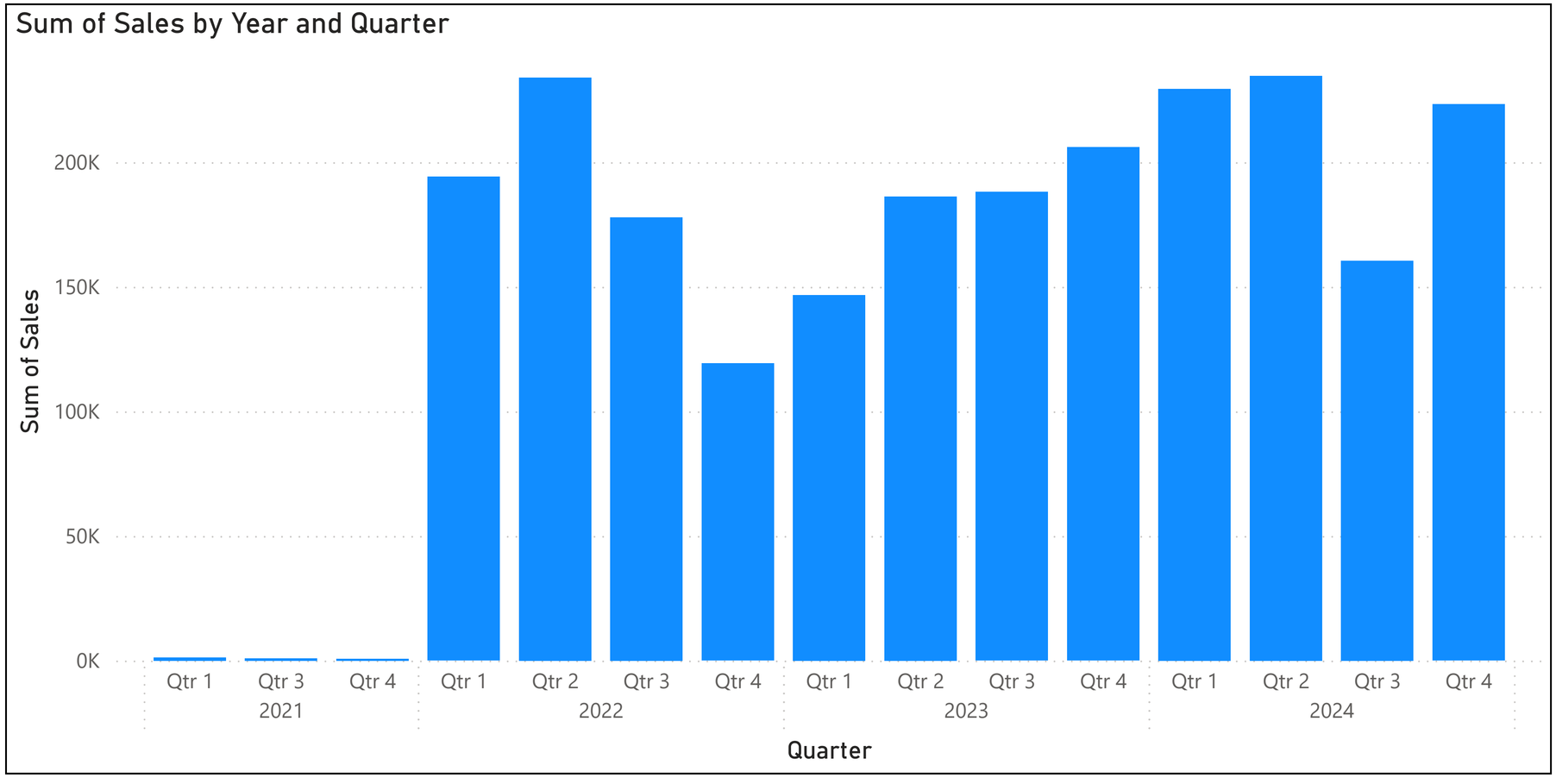
Drill-Down allows users to move from a higher level of data to a more detailed level within the same visual.
It works with hierarchical data.
Example Hierarchy
Year → Quarter → Month → Day

Drill-Through in Power BI

Drill-Through allows users to navigate to another report page that shows detailed information about a selected value.
It moves from one page to another page.

Drill-Down in Power BI allows users to explore hierarchical data within the same visual by moving from higher-level data to more detailed levels. Drill-Through allows users to navigate to another report page that shows detailed information about a selected value. For example, a user can drill down from Year to Month in a sales chart, while drill-through allows the user to click a product and view a detailed page showing region-wise sales for that product.

Ques 11: Use the Drill-Down feature to analyze sales trends from
Year → Quarter → Month → Day



Ques 12: Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000.
Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column

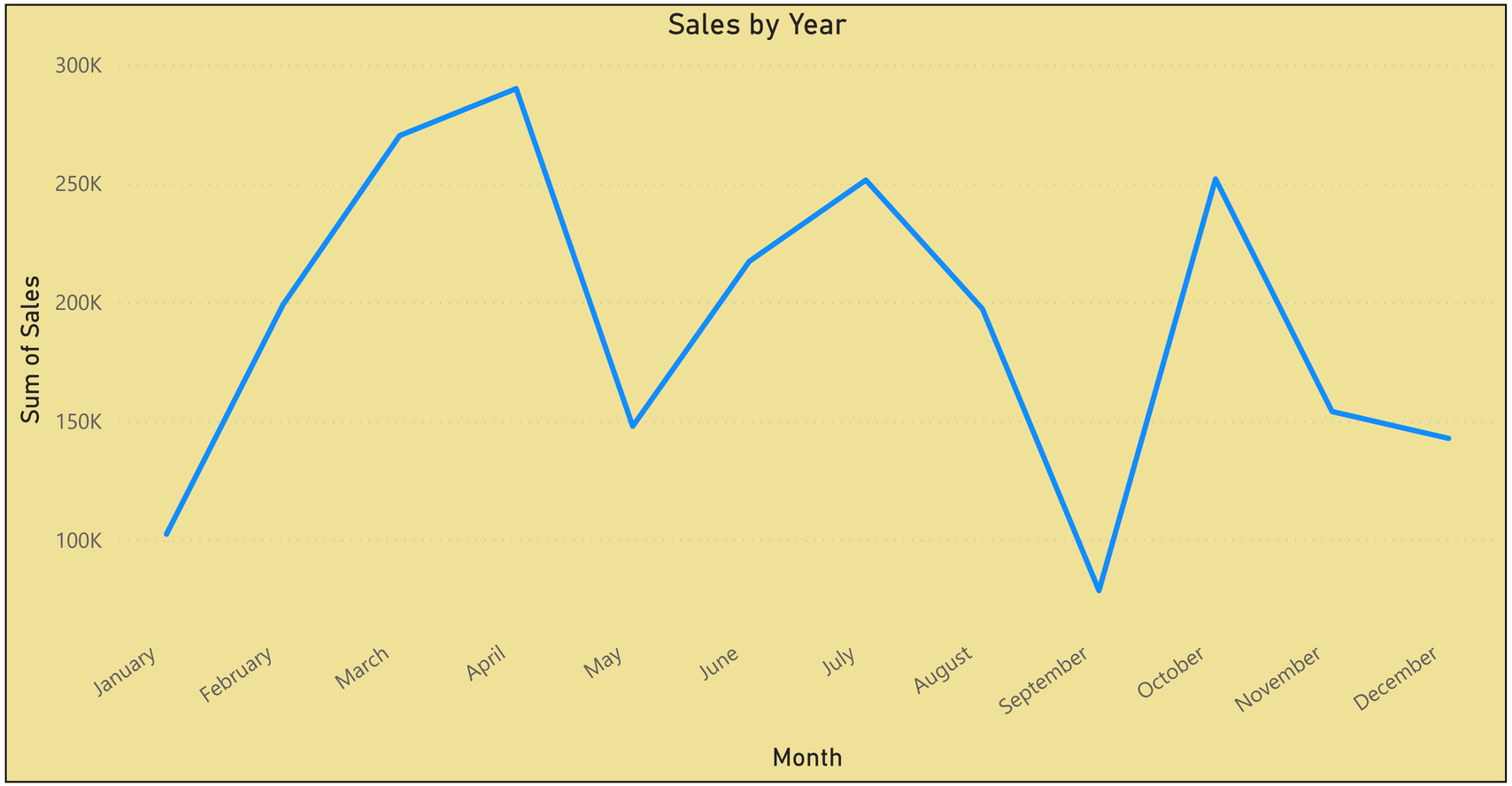
Product	Total Sales
Tablet	315818
Keyboard	304052
Printer	303220
Laptop	269552
Monitor	233383
Desk	231793
Phone	223560
Mouse	211644
Chair	208996
Total	**2302018**

Ques 13: What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction

Answer:

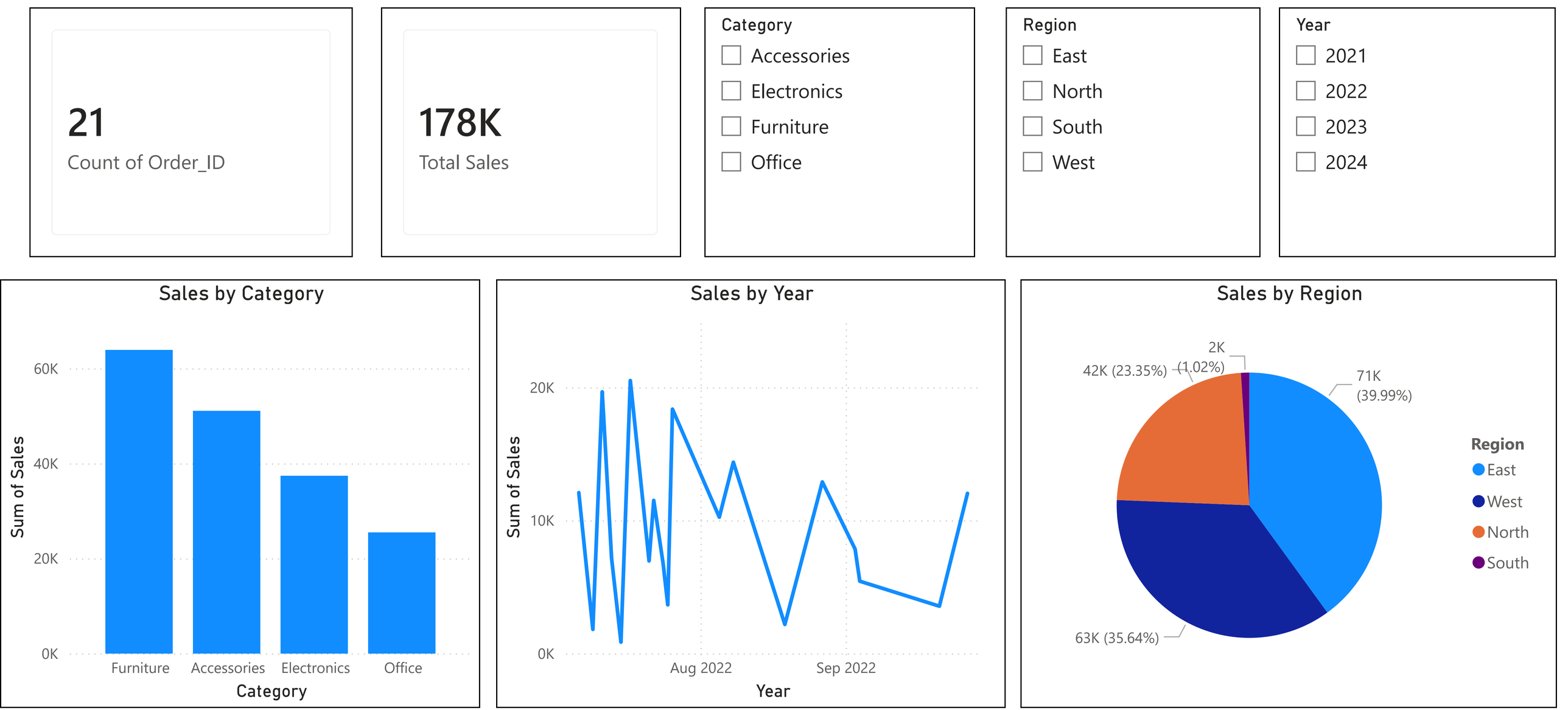
The “Keep All Filters” option in Drill-through determines whether the filters applied on the source page are carried over to the target page.
-When enabled, all existing filters (such as Region, Category, etc.) are retained along with the selected field.
-When disabled, only the selected drill-through field is passed to the target page.
This option improves user interaction by maintaining context and providing more relevant and consistent insights on the drill-through page.

Ques 14 : Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time



Year	Quarter	Month	Day	Sum of Sales
2021	Qtr 1	January	14	650
2021	Qtr 1	February	25	450
2021	Qtr 1	March	30	125
2021	Qtr 3	August	20	550
2021	Qtr 3	September	25	350
2021	Qtr 4	October	30	75
2022	Qtr 1	January	8	14678
2022	Qtr 1	January	10	19755
2022	Qtr 1	January	15	3728
2022	Qtr 1	January	25	10295
2022	Qtr 1	February	5	5280
2022	Qtr 1	February	6	12543
2022	Qtr 1	February	10	5335
2022	Qtr 1	February	16	18140
2022	Qtr 1	February	20	7696
2022	Qtr 1	February	23	12683
2022	Qtr 1	February	27	4127
2022	Qtr 1	March	3	9287
2022	Qtr 1	March	4	7470
2022	Qtr 1	March	16	9337
2022	Qtr 1	March	22	19064
2022	Qtr 1	March	27	17009
2022	Qtr 1	March	31	17660
2022	Qtr 2	April	1	15985
2022	Qtr 2	April	6	10003
2022	Qtr 2	April	9	19615
2022	Qtr 2	April	14	16086
2022	Qtr 2	April	15	7766
Total				**2302018**

Ques 15 : Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data.



Category	Sum of Sales
Accessories	749079
Electronics	808930
Furniture	440789
Office	303220
Total	**2302018**

Year	Sum of Sales
2021	2200
2022	725157
2023	726999
2024	847662
Total	**2302018**

Region	Sum of Sales
East	535112
North	550844
South	590265
West	625797
Total	**2302018**